

Course Announcements

- Coming Fri for this set of notes

Version Control: Practice

Note: You're encouraged to put projects on GitHub. Please do not put assignments on GitHub.

- Assignment 1: Part 1
 - This will get you practice with git & GitHub
 - Understand what you're doing in the assignment!
 - You may have to google, ask others, spend some time with this!
- you'll be using GitHub for your final projects
- [COGS 108 Resources - git](#)
 - If you use any of these and have feedback (good, bad, otherwise) please lmk
 - If you find other resources to add, feel free to suggest (maybe a PR?!)

COGS 108 Final Project

The COGS 108 Final Project will give you the chance to explore a topic of your choice and to expand your analytical skills. By working with real data of your choosing you can examine questions of particular interest to you.

- You are encouraged to work on a topic that matters to the you (you, your family, your neighborhood, your community, a state/province, country, community, etc).
- **Taboo Topics:** Movie Predictions/Recommendation System; YouTube Data Analysis, Kickstarter success prediction/analysis, prediction of what makes a song popular on Spotify & Whatever is MOST popular EVER and whatever is HOTTEST RN on Kaggle

Project Objectives

- Identify the problems and goals of a *real* situation and dataset.
- Choose an appropriate approach for formalizing and testing the problems and goals, and be able to articulate the reasoning for that selection.
- Implement your analysis choices on the dataset(s).
- Interpret the results of the analyses.
- Contextualize those results within a greater scientific and social context, acknowledging and addressing any potential issues related to privacy and ethics.
- Work effectively to manage a project as part of a team.

Finding groupmates

- Go to discussion lab next week!
- Chat with people in lecture, post on Piazza
 - Include: topic(s) interested in, section availability, lecture availability, how you'd like to communicate, how regular to meet/schedule, meeting modality, what your strengths/weaknesses are

Important group mate considerations:

- Topic
- Schedules
- Meeting modality
- Diverse strengths

Data

Tidy Data & Data Intuition

Shannon E. Ellis, Ph.D.
UC San Diego

Department of Cognitive Science
sellis@ucsd.edu



Data Structures Review

Structured data

- can be stored in database SQL
- tables with rows and columns
- requires a relational key
- 5-10% of all data

Semi-structured data

- doesn't reside in a relational database
- has organizational properties (easier to analyze)
- CSV, XML, JSON

Unstructured

- non-tabular data
- 80% of the world's data
- images, text, audio, videos

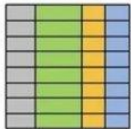
Structured Data (Databases)

- Manage huge datasets
- Control access
- Allow users to find a subset of data with a query
- Can run analysis on data and return a report

	Relational Databases (RDB)	Non-relational
Examples	SQLite, MySQL, Postgres	Hadoop, Hive, Apache CouchBase
How	Work using tables of data with “relationships” established between tables	Work with key-value pairs to lookup data (see JSON coming up)

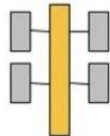
TYPES OF DATABASES

Relational



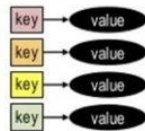
SQL

Analytical (OLAP)



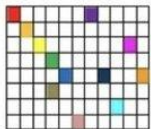
SQL

Key-Value



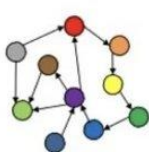
NOSQL

Column-Family



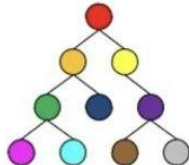
NOSQL

Graph



NOSQL

Document

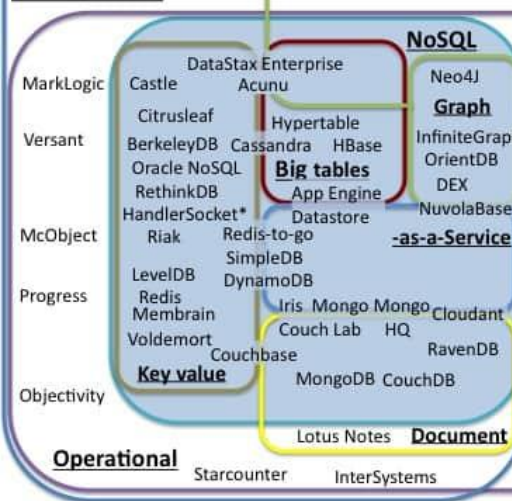


NOSQL

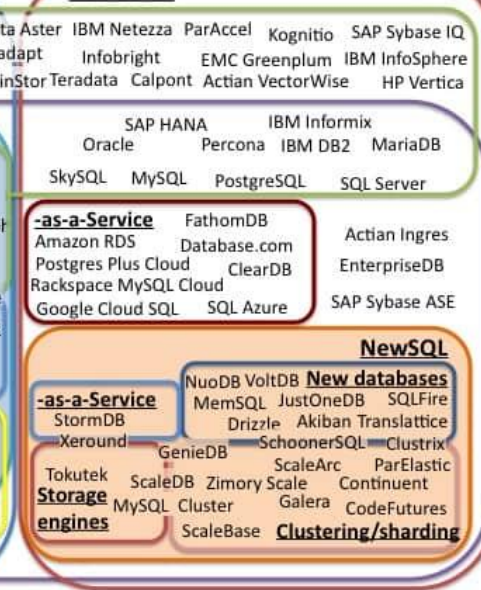
The evolving database landscape

451 Research

Non-relational



Relational



How do we get data from a RDB (Relational Database)?

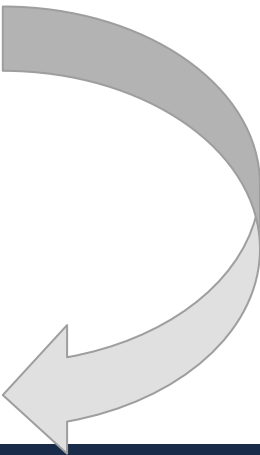
One of the simplest queries we can run!

A Query (SQL code)

```
SELECT population
FROM world
WHERE name = 'Germany'
```

Query Result

population
80716000



The 'world' Table in a DB

world				
name	continent	area	population	gdp
Afghanistan	Asia	652230	25500100	20343000000
Albania	Europe	28748	2831741	12960000000
Algeria	Africa	2381741	37100000	188681000000
Andorra	Europe	468	78115	3712000000
Angola	Africa	1246700	20609294	100990000000
....				

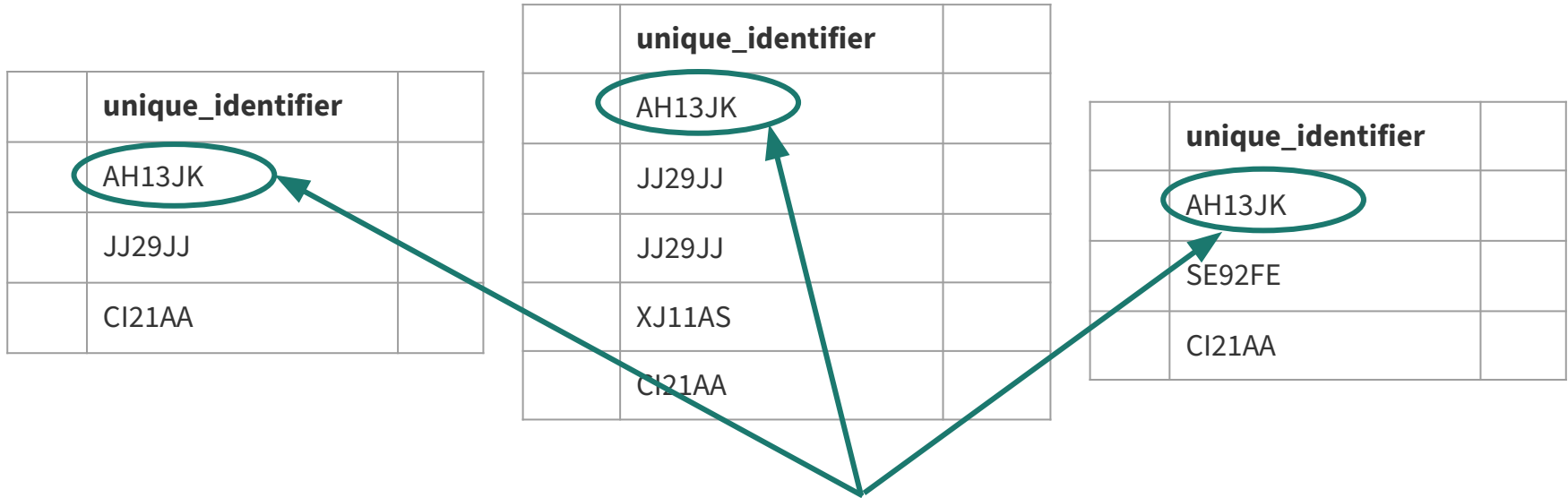
Relational Databases: A set of interdependent tables

1. Efficient Data Storage
2. Avoid Ambiguity
3. Increase Data Privacy

Results		Messages		
	EmployeeId	FirstName	LastName	DepartmentName
1	1	Ken	Sanchez	Executive
2	2	Teri	Duffy	Engineering
3	3	Roberto	Tamburello	Engineering
4	4	Rob	Walters	Engineering
5	5	Gail	Erickson	Engineering
6	6	Jossef	Goldberg	Engineering
7	7	Dylan	Miller	Support
8	8	Diane	Margheim	Support
9	9	Gigi	Matthew	Support
10	10	Michael	Raheem	Support

relational database

Information is stored across tables



entries are *related* to one another by their unique identifier

relational database

restaurant

name	id	address	type
Taco Stand	AH13JK	1 Main St.	Mexican
Pho Place	JJ29JJ	192 Street Rd.	Vietnamese
Taco Stand	XJ11AS	18 W. East St.	Fusion
Pizza Heaven	CI21AA	711 K Ave.	Italian

health inspections

id	inspection_date	inspector	score
AH13JK	2018-08-21	Sheila	97
JJ29JJ	2018-03-12	D'eonte	98
JJ29JJ	2018-01-02	Monica	66
XJ11AS	2018-12-16	Mark	43
CI21AA	2018-08-21	Anh	99

rating

id	stars
AH13JK	4.9
JJ29JJ	4.8
XJ11AS	4.2
CI21AA	4.7

relational database

restaurant

name	id	address	type
Taco Stand	AH13JK	1 Main St.	Mexican
Pho Place	JJ29JJ	192 Street Rd.	Vietnamese
Taco Stand	XJ11AS	18 W. East St.	Fusion
Pizza Heaven	CI21AA	711 K Ave.	Italian

health inspections

id	inspection_date	inspector	score
AH13JK	2018-08-21	Sheila	97
JJ29JJ	2018-03-12	D'eonte	98
JJ29JJ	2018-01-02	Monica	66
XJ11AS	2018-12-16	Mark	43
CI21AA	2018-08-21	Anh	99

rating

id	stars
AH13JK	4.9
JJ29JJ	4.8
XJ11AS	4.2
CI21AA	4.7

Two different restaurants with
the same name will have
different unique identifiers

relational database

Semi-structured Data (CSV, JSON, XML)



Example CSV



File Edit View Insert Format Data Tools Add-ons Help [All changes saved in Drive](#)

100% \$ % .0 .00 123 Arial 10 B I A

fx

	A	B	C	D	E	F
1	Email	First Name	Last Name	Company	Snippet 1	
2	example1@domain.com	John	Smith	Company 1	Snippet Sentence1	
3	example2@gmail.com	Mary	Blake	Company 2	Snippet Sentence 2	
4	example3@outlook.com	James	Joyce	Company 3	Snippet Sentence 3	
5						
6						
7						
8						

CSV file



Example CSV - Sheet1 — Notatnik

Plik Edycja Format Widok Pomoc

```
Email,First Name,Last Name,Company,Snippet 1
example1@domain.com,John,Smith,Company 1,Snippet Sentence1
example2@gmail.com,Mary,Blake,Company 2,Snippet Sentence 2
example3@outlook.com,James,Joyce,Company 3,Snippet Sentence 3
```

JSON: key-value pairs

nested/hierarchical data

```
{"Name": "Isabela"}
```

key



value

JSON

These are all
nested within
attributes

These are all
nested within
"Good For"

```
"attributes": {  
  "Take-out": true,  
  "Wi-Fi": "free",  
  "Drive-Thru": true,  
  "Good For": {  
    "dessert": false,  
    "latenight": false,  
    "lunch": false,  
    "dinner": false,  
    "breakfast": false,  
    "brunch": false  
  },  
}
```

JSON

```
{
  "cells": [
    {
      "cell_type": "markdown",
      "metadata": {},
      "source": [
        "This example represents the output the t-SNE dimensionality reduction algorithm on embeddings computed from Unicode emojis using Keras"
      ]
    },
    {
      "cell_type": "code",
      "execution_count": null,
      "metadata": {},
      "outputs": [],
      "source": [
        "import pandas as pd\n",
        "import holoviews as hv\n",
        "hv.extension('bokeh')\n"
      ]
    },
    {
      "cell_type": "markdown",
      "metadata": {},
      "source": [
        "## Declaring data"
      ]
    },
    {
      "cell_type": "code",
      "execution_count": null,
```



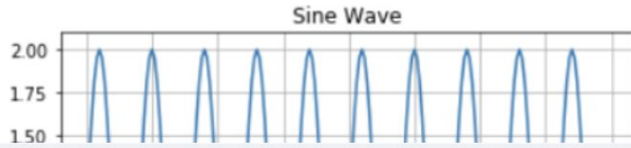
```
In [10]: import numpy as np
import matplotlib.pyplot as plt

# Data for plotting
t = np.arange(0.0, 2.0, 0.01)
s = 1 + np.sin((5 * 2)* np.pi * t)

# Note that using plt.subplots below is equivalent to using
# fig = plt.figure() and then ax = fig.add_subplot(111)
fig, ax = plt.subplots()
ax.plot(t, s)

ax.set(xlabel='time (s)', ylabel='voltage (mV)', title='Sine Wave')
ax.grid()
```

Out[10]:



"outputs": [

{

"data": {

"image/png":

"iVBORw0KGgoAAAANSUhEUgAAAYwAAAEWCAYAAAB1xKBvAAAABHNCSVQICAgIfAhkiAAAAAwSFlzAAALEgAACxIB0t1+/AAAADl
 0RVh0U29mdHdhcmUAAbWF0cGxvdGxpYiB2ZXJzaW9uIDIuMi4yLCBodHRwOi8vbWF0cGxvdGxpYi5vcmcvhp/UCwAAIABJREFUeJz
 svXmcHNd13/s9vc4+2EgABHeQEKVVSXGGRFLembFNSPn7Wyy45i5UXh5ZjvcSy4xcr78WK5bwkzvKSeIll0qaVxZKc0JLN+FHc0dx
 JEVxAAGQBAiCiIdbDP0tPT+80fVdXdmOnl1q17ezBm/T6f+QDdXVXnVtU996z3HFFKESNGjBgxYvRDYrkHECNGjBgxVgZigREjRow
 YMbQQC4wYMWLEiKGFWDGEiBEjRgwtxAIjRowYMWJoIRYYMWLEiBFDC7HAiBEDEJG/JiKPL/c4YsQ4nxELjBgfGojIXSLyoojMiMg

Jupyter Notebooks suck to be version controlled...

Clear Output Manually

The simplest solution is to always clear the output before committing. **Cell** → **All Output** → **Clear** → **Save**. This removes any binary blobs that have been generated by the notebook. There are three main drawbacks:

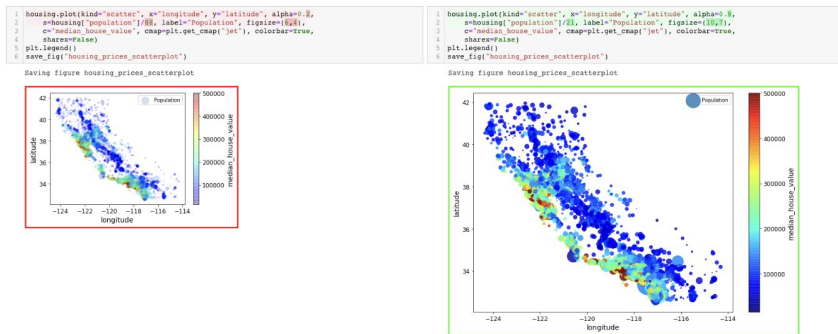
- It is a manual process.
- Collaborators on other machines will need to rerun the notebook to see the output, requiring additional time and setup.
- Collaborators on other machines may still create noise when new metadata is generated, like this information at the end of `simple-nb.ipynb`:

Version controlling notebooks sucks rocks

<https://nextjournal.com/schmudde/how-to-version-control-jupyter>

ReviewNB

ReviewNB is a GitHub app that also offers visual diffing with an interface that looks similar to the traditional Jupyter IDE. Because the outputs are visualized, problems associated with committing binary blobs disappear.



ReviewNB example courtesy of the [ReviewNB website](#)



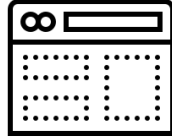
Unstructured Data

Some datasets record information about the state of the world, but in a more heterogeneous way. Perhaps it is a large text corpus with images and links like Wikipedia, or the complicated mix of notes and test results appearing in personal medical records.

Unstructured Data Types



Text files and
documents



Websites and
applications



Sensor
data



Image
files



Audio
files



Video
files



Email
data



Social media
data



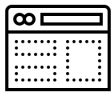
Positive:
70%

Negative:
20%

Neutral:
10%



Text:
Sentiment
Analysis



PYTHON

BEAUTIFULSOUP WEB SCRAPING





Bedroom Or Not?



“The left two photos were correctly predicted as bedrooms; The right two photos were correctly predicted NOT as bedrooms.”

Tidy Data

"Good data scientists understand, in a deep way, that the heavy lifting of cleanup and preparation isn't something that gets in the way of solving the problem: it is the problem."

- DJ Patil

untidy data

Australian Bureau of Statistics												
1800.0 Australian Marriage Law Postal Survey, 2017												
Released on 15 November 2017												
Table 5 Participation by Federal Electoral Division(a), Males and Age Gender apartheid												
Yeah NA												
Total participants												
Eligible participants												
Participation rate (%)												
Merged cells												
Total participants												
Eligible participants												
Participation rate (%)												
Northern Territory												
Total participants												
Eligible participants												
Participation rate (%)												
Australian Capital Territory Divisions												
Total participants												
Eligible participants												
Participation rate (%)												
Fenner(e)												
Total participants												
Eligible participants												
Participation rate (%)												
NA Yeah												
Total participants												
Eligible participants												
Participation rate (%)												
Australia												
Total participants												
Eligible participants												
Participation rate (%)												
a) The Federal Electoral Divisions are current as at 24 August 2017												
b) Includes those whose age is unknown												
c) Includes Christmas Island and the Cocos (Keeling) Islands												
d) Includes Norfolk Island												
e) Includes Jarvis Bay												

tidy data

1	area	gender	age	State	Area (sq km)	Eligible participants	Participation rate (%)	Total participants	Total Participants
2	Adelaide	Female	18-19 years	SA	76	1341	83.5	1120	1120
3	Adelaide	Female	20-24 years	SA	76	4620	81.2	3750	3750
4	Adelaide	Female	25-29 years	SA	76	4897	81.8	4004	4004
5	Adelaide	Female	30-34 years	SA	76	4784	79.8	3820	3820
6	Adelaide	Female	35-39 years	SA	76	4319	79	3411	3411
7	Adelaide	Female	40-44 years	SA	76	4310	80.6	3472	3472
8	Adelaide	Female	45-49 years	SA	76	4579	81.4	3728	3728
9	Adelaide	Female	50-54 years	SA	76	4475	84.7	3791	3791
10	Adelaide	Female	55-59 years	SA	76	4622	87.3	4033	4033
11	Adelaide	Female	60-64 years	SA	76	4342	89.3	3879	3879
12	Adelaide	Female	65-69 years	SA	76	3970	90.7	3602	3602
13	Adelaide	Female	70-74 years	SA	76	3009	90.3	2716	2716
14	Adelaide	Female	75-79 years	SA	76	2156	88.5	1908	1908
15	Adelaide	Female	80-84 years	SA	76	1673	85.1	1423	1423

data
→
wrangling

Tidy Data

1. Each **variable** you measure should be in a single column

	A	B	C	D	E	F	G
1	ID	LastName	FirstName	Sex	City	State	Occupation
2	1004	Smith	Jane	female	Frederick	MD	Welder
3	4587	Nayef	Mohammed	male	Upper Darby	PA	Nurse
4	1727	Doe	Janice	female	San Diego	CA	Doctor
5	6879	Jordan	Alex	male	Birmingham	AL	Teacher

2. Every **observation** of a variable should be in a different row

	A	B	C	D	E	F	G
1	ID	LastName	FirstName	Sex	City	State	Occupation
2	1004	Smith	Jane	female	Frederick	MD	Welder
3	4587	Nayef	Mohammed	male	Upper Darby	PA	Nurse
4	1727	Doe	Janice	female	San Diego	CA	Doctor
5	6879	Jordan	Alex	male	Birmingham	AL	Teacher

3. There should be one table for each type of data

Demographic Survey Data

	A	B	C	D	E	F	G
1	ID	LastName	FirstName	Sex	City	State	Occupation
2	1004	Smith	Jane	female	Frederick	MD	Welder
3	4587	Nayef	Mohammed	male	Upper Darby	PA	Nurse
4	1727	Doe	Janice	female	San Diego	CA	Doctor
5	6879	Jordan	Alex	male	Birmingham	AL	Teacher

Doctor's Office Measurements Data

	A	D	E	F	G
1	ID	Height_inches	Weight_lbs	Insulin	Glucose
2	1004	65	180	0.60	163
3	4587	75	215	1.46	150
4	1727	62	124	0.72	177
5	6879	77	160	1.23	205

4. If you have multiple tables, they should include a column in each *with the same column label* that allows them to be joined or merged

	A	B	C	D	E	F	G
1	ID	LastName	FirstName	Sex	City	State	Occupation
2	1004	Smith	Jane	female	Frederick	MD	Welder
3	4587	Nayef	Mohammed	male	Upper Darby	PA	Nurse
4	1727	Doe	Janice	female	San Diego	CA	Doctor
5	6879	Jordan	Alex	male	Birmingham	AL	Teacher

	A	D	E	F	G
1	ID	Height_inches	Weight_lbs	Insulin	Glucose
2	1004	65	180	0.60	163
3	4587	75	215	1.46	150
4	1727	62	124	0.72	177
5	6879	77	160	1.23	205

Tidy data == rectangular data

A

	A	B	C	D	E
1	id	sex	glucose	insulin	triglyc
2	101	Male	134.1	0.60	273.4
3	102	Female	120.0	1.18	243.6
4	103	Male	124.8	1.23	297.6
5	104	Male	83.1	1.16	142.4
6	105	Male	105.2	0.73	215.7

Tidy Data Benefits

1. consistent data structure
2. foster tool development
3. require only a small set of tools to be learned
4. allow for datasets to be combined

TIDY data is NOT the same as CLEAN data

Clean Data

Data cleaning is the process of detecting and correcting/removing data records that are:

- incomplete,
- incorrect,
- inaccurate, or
- irrelevant

...and often includes making sure missing data is correctly marked

...and that duplicate records and other errors are removed

Data that are **TIDY** and **CLEAN** are ready for use

Tabular Data Time

<https://forms.gle/DFnkirtqVDB7Y73r8>



A

ID	Last	First	height_m	height_f
1004	Smith	Jane	NA	65
4587	Nayef	Mohammed	72	NA
1727	Doe	Janice	NA	60
6879	Jordan	Alex	55	NA

B

ID	Last	First	height_m	height_f
1004	Smith	Jane		65
4587	Nayef	Mohammed	72	
1727	Doe	Janice		60
6879	Jordan	Alex	55	

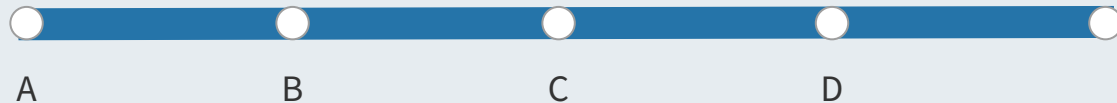
C

ID	Last	First	sex	height
1004	Smith	Jane	female	65
4587	Nayef	Mohammed	male	72
1727	Doe	Janice	fem	60
6879	Jordan	Alex	male	55

D

ID	Last	First	sex	height
1004	Smith	Jane	F	65
4587	Nayef	Mohammed	M	72
1727	Doe	Janice	F	60
6879	Jordan	Alex	M	55

Which of these tables stores data best?



A

B

C

D

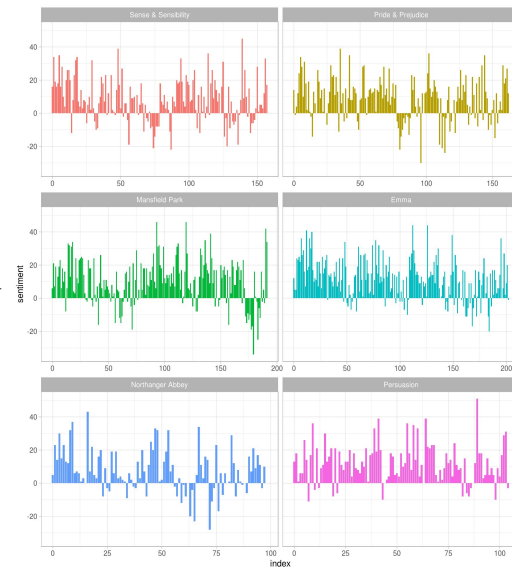


tidy dataset

Word	Novel	Frequency
good	Emma	359
young	Emma	192
friend	Emma	166



results





	date	lie	explanation	url
0	Jan. 21, 2017	I wasn't a fan of Iraq. I didn't want to go in.	He was for an invasion before he was against it.	https://www.buzzfeed.com/andrewkaczynski/in/2017/01/21/06/10-things-you-didnt-know-about-trump
1	Jan. 21, 2017	A reporter for Time magazine — and I have been.	Trump was on the cover 11 times and Nixon appt...	http://nation.time.com/2017/01/21/06/10-things-you-didnt-know-about-trump
2	Jan. 23, 2017	Between 3 million and 5 million illegal votes	There's no evidence of illegal voting.	https://www.nytimes.com/2017/01/23/us/politics.
3	Jan. 25, 2017	Now, the audience was the biggest ever. But th...	Official aerial photos show Obama's 2009 inaug...	https://www.nytimes.com/2017/01/21/us/politics.
4	Jan. 25, 2017	Take a look at the Pew reports (which show vot...	The report never mentioned voter fraud.	https://www.nytimes.com/2017/01/24/us/politics.

results

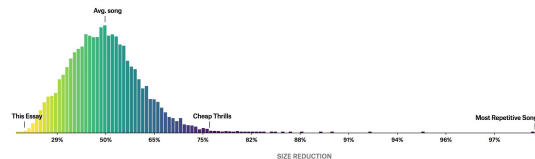
First day without
a public lie
or falsehood MA

 text (lyrics)

tidy dataset

song	Artist	Released	Reduction
Cheap Thrills	Sia	2016	76
Around The World	Daft Punk	1997	98
Everybody Dies	J. Cole	2018	27

results



Data Intuition



1011



1375



In today's pattern recognition class my professor talked about PCA, eigenvectors and eigenvalues.

I understood the mathematics of it. If I'm asked to find eigenvalues etc. I'll do it correctly like a machine. But I didn't **understand** it. I didn't get the purpose of it. I didn't get the feel of it.

I strongly believe in the following quote:

You do not really understand something unless you can explain it to your grandmother. -- Albert Einstein

Well, I can't explain these concepts to a layman or grandma.

1. Why PCA, eigenvectors & eigenvalues? What was the *need* for these concepts?
2. How would you explain these to a layman?

Theory vs. Practice: “Tai’s model”

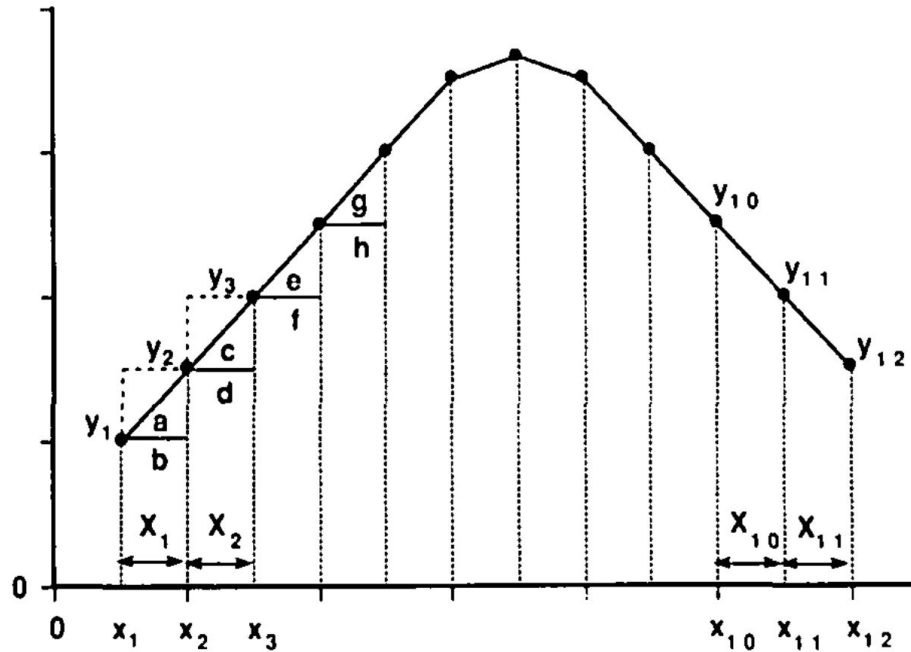
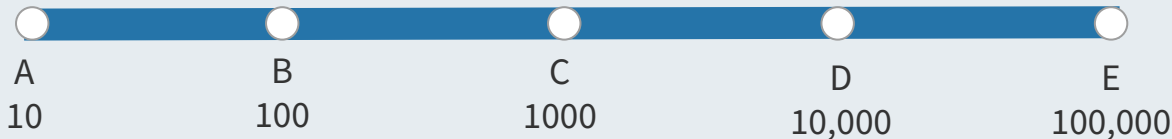


Figure 1—Total area under the curve is the sum of individual areas of triangles a, c, e, and g and rectangles b, d, f, and h.

Fermi Estimation



Approximately how many piano tuners do you think there are in the city of Chicago?

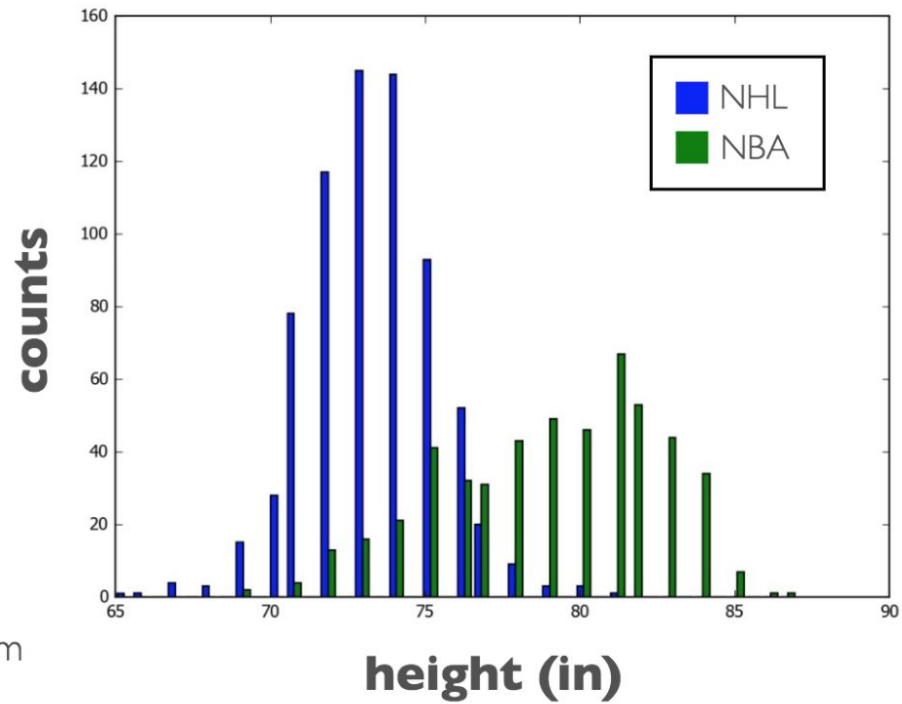
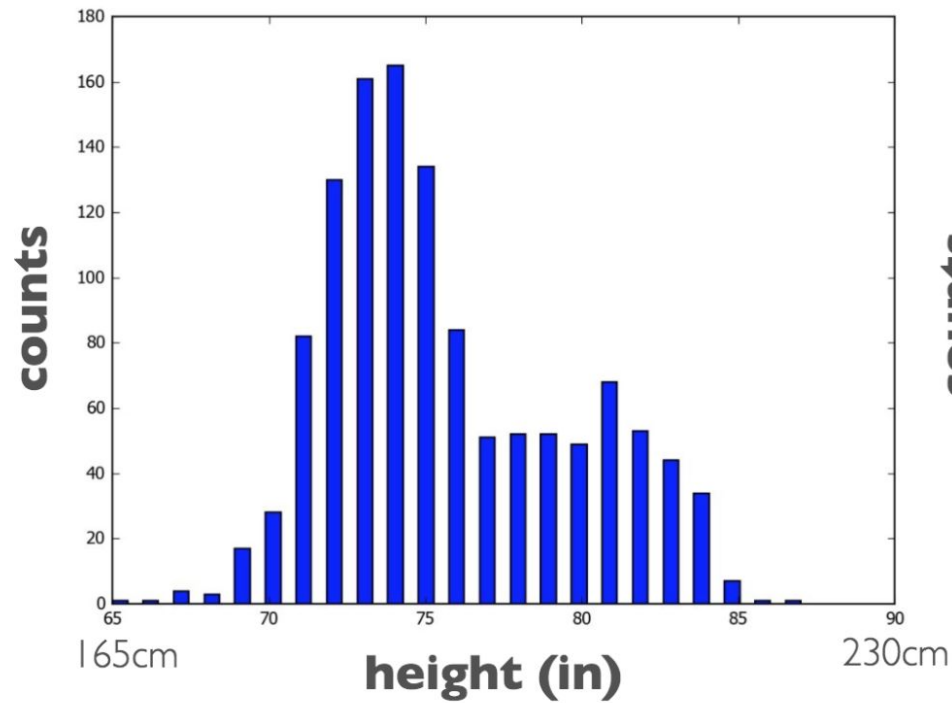


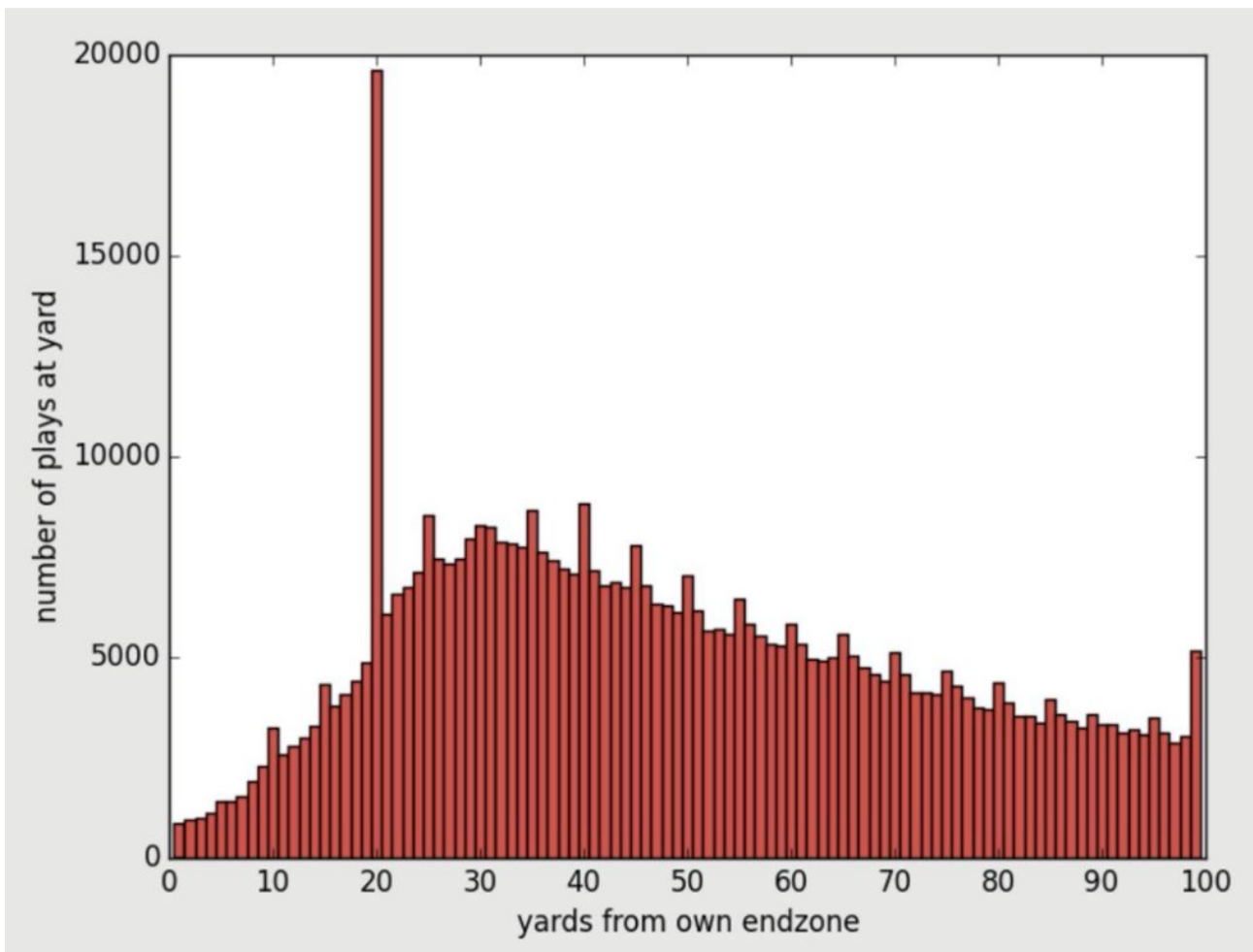


Fermi: Think. Pair. Share.



**...how does data intuition matter in
data science?**

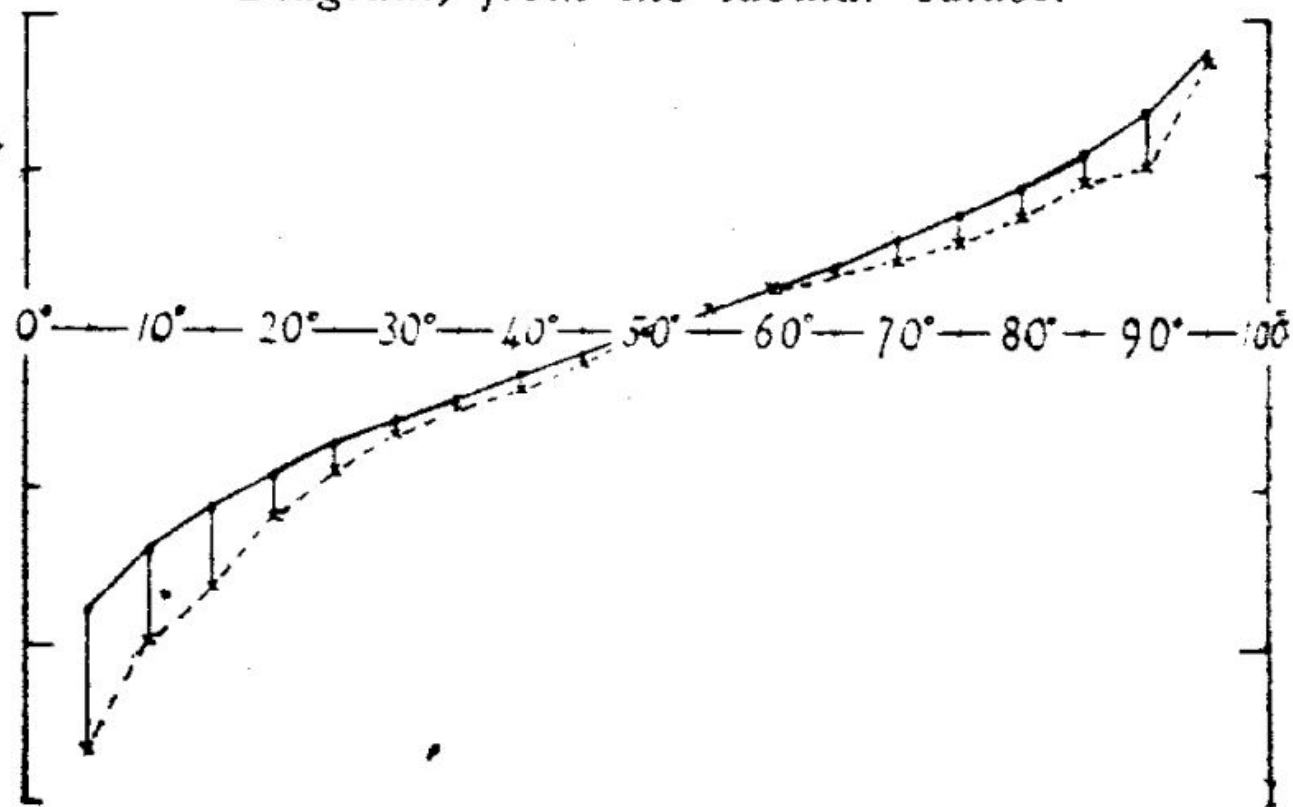




Data Intuition

1. Think about your question and your expectations
2. Do some Fermi calculations (back of the envelope calculations)
3. Write code & look at outputs <- think about those outputs
4. Use your gut instinct / background knowledge to guide you
5. Review code & fix bugs

Diagram, from the tabular values.



Vox Populi

The Wisdom of the Crowds

- Diversity of opinion: Each person should have private information....even if it's just an eccentric interpretation of the known facts
- Independence: People's opinions aren't determined by the opinions of those around them
- Decentralization: People are able to specialize and draw on local knowledge
- Aggregation: Some mechanism exists for turning private judgements into a collective decision

Wisdom of the Crowds...

(don't Google/GenAI or discuss with one another until after you submit)

